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A MEOL logic layout optimization recommendation under 3 nm CFET architecture and beyond

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Abstract

In order to continue the area shrink of the integrated circuit area, besides patterning technology improvement, an introduction of Complementary Field Effect Transistor (CFET) has been proposed for 3 nm logic node [1-2]. We have designed an SRAM under CFET architecture, where a N-type Metal Oxide Semiconductor (NMOS) is stacked on top of a PMOS. The method of local interconnect of two groups of MOSFETs (PMOS and NMOS) is critical to the area of this SRAM unit cell. We have used our self-developed Extreme Ultra-Violet (EUV) process simulation code with photon absorption stochastics [3-5] to study a 3 nm CFET SRAM processes. In addition, according to our previous study on the industry standard of the photolithographic process, we have incorporated the photon absorption stochastics and determined the EUV process window industry standard for the future technology nodes [6]. From the perspective of the industry standard, we have found that once the 45° local interconnect is adopted, this SRAM unit cell can reach a minimum area of ~ 0.0115μm^2. However, if we keep the original square local interconnect design, the cell height needs to be increased from 5 fin pitches (FPs) to at least 6 FPs with the SRAM unit cell area increasing by at least 20%.

In this paper, we will study the process window of the original square local interconnect designs with stitching in both low and high NA EUV lithography and the corresponding minimum SRAM area.

1. Introduction

Our previous papers indicate that the optimum pitch for the contact holes is around 48 ~ 50 nm under 0.33 Numerical Aperture (NA) Extreme Ultra-Violet (EUV) single exposure process [7]. From the simulation results, we have found 0.33 NA EUV can still be used to complete the lithography patterning in the 3 nm logic node. In addition, high NA EUV exposure machines can first be introduced into the 3 nm node process to get "warmed up" before applied to more advanced logic technology nodes, such as 2 nm, 1 nm nodes [8]. According the International Roadmap for Devices and Systems (IRDS) roadmap, the minimum pitches of via layers will ultimately be 20 nm. Therefore, the anchor pitch for the via is around 38 ~ 40 nm in 0.55 NA EUV lithography in this paper.

We have described the algorithm and multi-layer mask film stack used by our self-developed simulation program. Compared with the 6° of 0.33 NA, the Chief Ray Angle at Object space (CRAO) of incidence in 0.55 NA EUV lithography is 5.355°. High NA EUV lithography uses the anamorphic 8:1 and 4:1 reduction ratios along the scanning direction (Y-axis) and perpendicular direction (X-axis), respectively. While low NA EUV lithography uses the same 4:1 reduction ratios along both Y- and X-axes [8].

As we all know, in order to realize a smaller SRAM unit cell, the Complementary Field Effect Transistor (CFET) architecture has been introduced to semiconductor chip manufacturing since the 3 nm logic node. In our previous papers, we have shown a tentative 3 nm CFET SRAM Layout and the corresponding flow which involve 45° local interconnect patterns. These patterns can be realized by both low and high NA EUV processes. However, in high EUV lithography, the process window is higher and the etch bias is smaller than those in low NA EUV.

In this paper, we will study the original square interconnect design including two adjacent L-shaped masks which need to be disassembled into three masks, and one of them needs to be stitched after two exposures in both low and high NA EUV processes.

2. Results and Discussion

2.1 A 3 nm CFET SRAM with different local interconnect design patterns

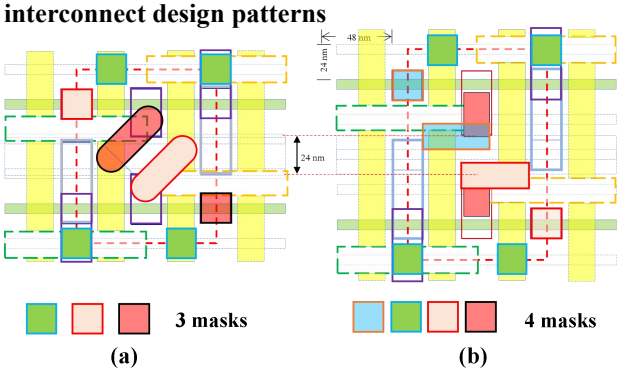


Figure 1. The M0G layer design (a) A 3 nm CFET SRAM with 45° local interconnect design patterns; (b) A 3 nm CFET SRAM with L-shaped local interconnect design patterns.

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Figure 1 shows a tentative 3 nm CFET SRAM layouts with different local interconnect design patterns. From the figure, the Fin Pitch (FP) is 24 nm, a contact-to-poly pitch (CPP) is 48 nm, and the minimum contact pitch that can be produced by 0.33 NA EUV is 48 nm. From Figure 1 (a), this M0G layer in Middle-End-Of-Line (MEOL) can be divided into three masks, two of which involve a 45° local connect design pattern. Owing to this 45° local interconnect design in both low and high NA EUV processes, the SRAM cell height only consists of 5 Fin Pitches (FPs) and the M0G layer contains 3 masks. However, if we keep the original square local interconnect design, as shown in Figure 1 (b), the cell height needs to be increased from 5 FPs to at least 6 FPs with the SRAM unit cell area increasing by at least 20%. From Figure 1 (b), two adjacent L-shaped masks need to be disassembled into three masks. In addition, two short bars that are perpendicular to each other in one L-shaped pattern need to be stitched together by two exposures. In other words, if we adopt the original square local interconnect design, not only the area of the SRAM need to be increased, but also the M0G layer need add one more mask.

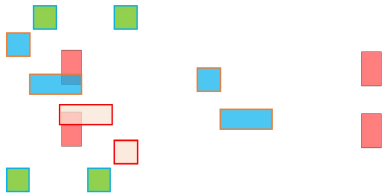


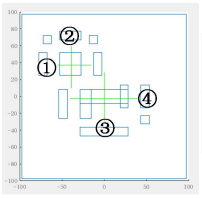
Figure 2. (a)The M0G layer need to be divided into 4 masks; the two most important masks are (b) “A” pattern and (c) “B” pattern.

In this paper, we mainly analyze the process windows of two of the four masks in M0G layer mentioned in Figure 1 (b) or Figure 2 (a) in both low and high NA EUV lithography processes. One of them is “A” pattern, which contains one short bar along X direction and one contact hole, as shown in Figure 2 (b). Another is “B” pattern, which contains two short bars along Y direction, as shown in Figure 2 (c).

2.2 The simulation results of “A” and “B” patterns in low NA EUV lithography

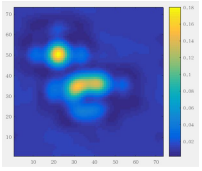
We have mentioned many times in our previous papers that taking stochastics effect into account, EL of most patterns is recommended to be at least 18% in EUV lithography [4]. The illumination condition of M0G layer in low NA EUV is an annular with 0.65-0.45 partial coherence factor and 0.33 NA. The Photo Resist (PR) is a typical 30 nm thick Chemically Amplified photoResist (CAR), which Effective Photoacid Diffusion Length (EPDL) is 3.5 nm. The anchor pattern of M0G layer is array dense contact holes with pitch of 48 nm. The After Developing Inspection (ADI) Critical Dimension (CD) is 24 nm and the EL is about 25% which is much larger than 18%.

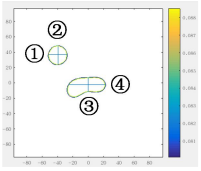
From the Figure 1 (b), we can find that the pitch of the two short bars (the short bar of “A” pattern) is 24 nm. In order to ensure that the space between these two short bars exceeds 10 nm, the maximum final etch CD along Y direction of these two short bars is 12 ~ 14 nm. The etching depth of this layer is shallow, therefore, the suitable ADI CD along Y direction of these two short bars is 18 nm with ~ 2 to 3 nm etch bias on each side. These two short bars need to be disassembled into two masks and matched with a contact hole respectively, i.e. the “A” pattern, as shown in Figure 2 (b). The mask, simulation aerial and contour images, and simulation results of “A” pattern in low NA EUV are shown in Figure 3.



Low NA A pattern				
Line cut	①	②	③	④
CD (nm)	23.8	24.1	18	48
EL	28.1%	30.3%	18.1%	41.7%

Mask





Aerial image Contour image

Figure 3. The mask, simulation aerial and contour images, and simulation results of “A” pattern in low NA EUV lithography.

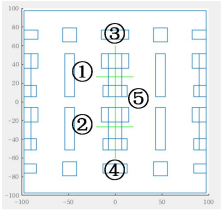
As mentioned above, the maximum ADI CD of line cut ③ is 18 nm. From the simulation results shown in Figure 3, when the ADI CD of the line cut ③ achieves 18 nm, the EL is about 18% which just meets the process requirements. Line cuts ① and ② are the contact hole patterns with the ADI CDs of ~ 24 nm and the ELs of ~ 28%, which are much greater than 18%.

The second pattern is “B” pattern, as shown in Figure 2 (c), the corresponding mask, simulation aerial and contour images, and simulation results in low NA EUV are shown in Figure 4. In order to obtain enough lithography process window of short bars and the minimum Tip to Tip (TtT) CD， the ADI CDs at line cut ① and ② is set to 20 nm. As documented previously [4], line cut ⑤ is the TtT pattern whose EL must be ≥ 13%. From the simulation results, we can find that, the minimum TtT CD is about 16 nm between these two short bars with the EL larger than 13% (14.6%).

As mentioned above, by optimizing the EPDL of PR and the illumination conditions, the original square interconnect designs which include stitching process can

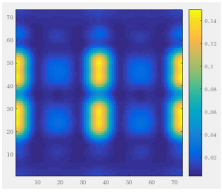
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be realized by low NA EUV, and the lithography process windows can just meet the requirement of mass production.

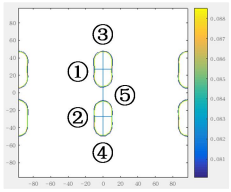


Mask

Low NA B pattern			
Line cut	①	②	
CD (nm)	19.8	20.1	
EL	19.9%	20.6%	
Line cut	③	④	⑤
CD (nm)	40.5	40	16.1
EL	40.6%	37.0%	14.6%



Aerial image



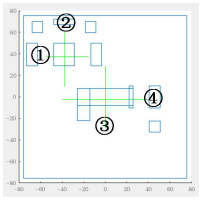
Contour image

Figure 4. The mask, simulation aerial and contour images, and simulation results of "B" pattern in low NA EUV lithography.

The smaller the TtT CD, the larger an overlap area of two short bars along X and Y directions between "A" and "B" patterns, respectively. Therefore, the overlay performance can be loosened accordingly, which is beneficial to the Research and Development (R&D) of lithography processes. In order to get smaller TtT CD, the CD (20 nm) at line cuts ① and ② shown in Figure 4 need to be slightly larger than that (18 nm) of at line cut ③ shown in Figure 3.

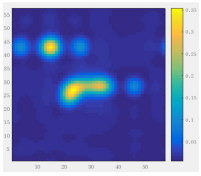
2.3 The simulation results of "A" and "B" patterns in high NA EUV lithography

The illumination condition of M0G layer in high NA EUV is an annular with 0.5-0.2 partial coherence factor and 0.55 NA. The PR is a typical 30 nm thick CAR, which EPDL is 3.5 nm. The anchor pattern of M0G layer is array dense contact holes with pitch of 38 nm in high NA EUV. The ADI CD is 18 nm and the EL is about 30% which is much larger than 18%.

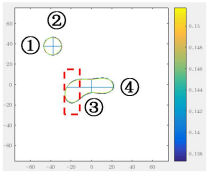


Mask

High NA A pattern				
Line cut	①	②	③	④
CD (nm)	17.4	17.4	13	46
EL	31.6%	31.6%	19.5%	70.5%



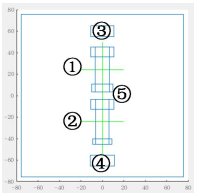
Aerial image



Contour image

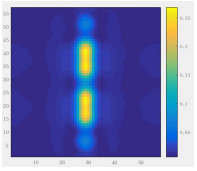
Figure 5. The mask, simulation aerial and contour images, and simulation results of "A" pattern in high NA EUV lithography.

The mask, simulation aerial and contour images, and simulation results of "A" pattern in high NA EUV are shown in Figure 5. When the ADI CD of the line cut ③ can be shrunk to 13 nm, the EL (19.5%) can still be greater than 18%. Line cuts ① and ② are the contact hole patterns with the ADI CDs of ~ 17.5 nm and the ELs of ~ 31%. It can also be seen that the obvious turning pattern in the red dotted box can make the M0G layer better connected with the metal gate layer.

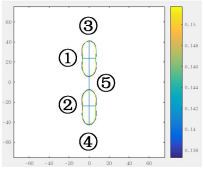


Mask

High NA B pattern			
Line cut	①	②	
CD (nm)	13	13	
EL	21.6%	21.6%	
Line cut	③	④	⑤
CD (nm)	35.2	36.7	13.1
EL	42.9%	39.8%	14.4%



Aerial image



Contour image

Figure 6. The mask, simulation aerial and contour images, and simulation results of "B" pattern in high NA EUV lithography.

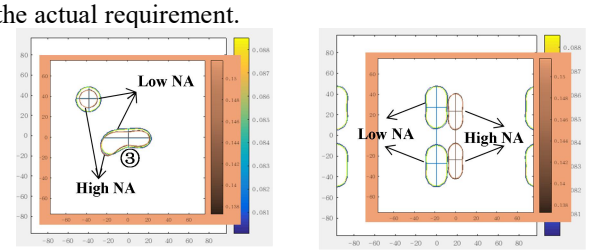
The mask, simulation aerial and contour images, and simulation results of "B" pattern in high NA EUV are shown in Figure 6. In order to have a same etch bias, between line cut ③ shown in Figure 5 and line cuts ① and ② shown in Figure 6, the ADI CDs of line cuts ① and ② are set to 13 nm which is same as that of line cut ③ shown in Figure 5.The ELs of line cuts ① and ② are much larger than 18%. As mentioned above, line cut ⑤ is the TtT pattern whose EL must be ≥ 13%. From the simulation results, the minimum TtT CD in high NA EUV can be reduced to ~ 13 nm with the EL larger than 13% (14.4%).

2.4 The comparison of simulation contours between low and high NA EUV lithography

Figure 7 (a) shows the simulation contours of "A" pattern under low and high NA EUV processes, which shows that the low NA EUV process requires a larger etch bias. If the final etch CD of line cut ③ is 12 nm, the etch bias in low NA EUV process is about 3 nm on each side, while the etch bias in high NA EUV process is only 0.5 nm on each side. In addition, as can be seen from Figure 5, even if the ADI CD at line cut ③ is reduced to 13 nm, the EL still exceeds 18% by ~ 1.5

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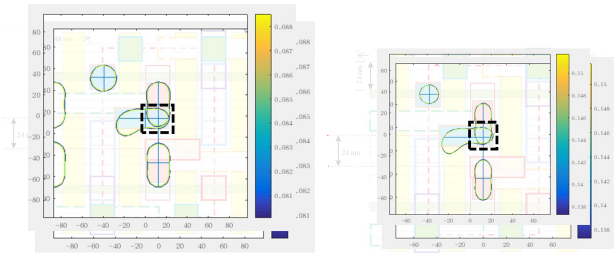
percentage points, so the ADI CD of line cut ③ in high NA EUV process can be continuously shrunk based on the actual requirement.



(a) (b)

Figure 7. The comparison of contour images between low and high NA EUV processes, (a) "A" patterns, (b) "B" patterns.

Figure 7 (b) shows the simulation contours of "B" pattern under low and high NA EUV processes. As can be seen from the figure, high NA EUV process can not only make the ADI CD smaller at line cuts ① and ②, but also make the two short bars closer along Y direction. If the final etch CD of line cuts ① and ② is 12 nm, the etch bias in low NA EUV process is about 4 nm on each side, while the etch bias in high NA EUV process is only 0.5 nm on each side. Similar to "A" pattern, even if the CDs of line cuts ① and ② has been reduced to 13 nm, the EL still exceeds 18% by ~ 3.5 percentage points, so the ADI CDs of line cuts ① and ② in high NA EUV process can also be continuously shrunk based on the actual requirement.



(a) (b)

Figure 8. The simulation results of "A" and "B" patterns in both low and high NA EUV are superimposed on the SRAM layout, respectively. (a) Low NA, (b) High NA contours.

Figure 8 shows the superposition of simulation contours both in low and high NA EUV and the 3 nm CFET SRAM layout. The black dotted box in the figure is the overlap area of the short bars which are perpendicular to each other in the "A" and "B" patterns, respectively. After etch process, the larger the overlap area, the better the reliability the local interconnect is. After etch, the overlap areas are about 50 nm^2 and 80 nm^2 under low and high NA EUV, respectively. Compared to the high NA EUV lithography, it is essential to more precisely control the overlay performance between "A" and "B" patterns in the low NA EUV process.

3. Summary

We have presented a 3 nm CFET SRAM Layout including 5 fin pitches and the corresponding flow which involve 45° local interconnect patterns in our previous papers. In this paper, we find that when the SRAM cell height is increased to 6 fin pitches, or the SRAM area is increased by 20%, the square local interconnect design can be used with the SRAM unit cell area increased from ~ 0.0115 μm^2 to ~ 0.0138 μm^2 and with the M0G layer, the traditional local interconnect layer, to need another mask compared to the 45° local interconnect design.

If we use the low NA EUV, there will be a larger etch bias for short bars with ~ 3 to 4 nm on each side for a final etch CD of 12 nm compared with the 0.5 nm etch bias in high NA EUV process. And the low NA process has smaller process windows compared with high NA process. Due to the large ADI CD of TtT pattern and large etch bias in the low NA EUV process, in order to obtain the maximum overlap area of two mutually perpendicular short bars, the overlay performance needs to be better than that in the high NA EUV process. Actually, the 3 nm logic node is the last one that can be realized by low NA EUV with good enough process window, and it is reasonable to expect that the high NA EUV process will be introduced in the subsequent 2 nm and 1 nm technology nodes.

Acknowledgments

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